

Semantic Uncertainty Propagation

A Self-Contained Theory of Individuation, Meaning,
and Comprehension-Free Translation in Finite Weighted Graphs

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Abstract

We develop, from a single derived fact, a complete theory of meaning and its transfer. The fact is a strictly positive *floor* $\beta > 0$ on the cost of telling one thing from the rest of a whole that cannot be finished: identifying a part is comparing it against everything it is not, and against a non-completable whole that comparison never closes, so an irreducible residue remains. Every object below is a finite weighted graph or a construction on one. From the floor we prove, as graph theorems: that a part is fixed only by its negation (*individuation by negation* admits no selector without regress); that the *identity* of a part is not a label but the invariant minimum cut that survives every relabelling, a region of weight $\geq \beta$ and never a point—so identity is the *least sufficient unknowability* of a thing, never the thing itself; that a *name* is the content-free handle on this cut—a stash of unknowability—and the *self*-name is necessarily late and approximate, the namer (having the largest complement) being the most expensive thing to individuate and forbidden by the diagonal of self-verification from naming itself directly, so that language begins at the individuated periphery and converges on the namer it never reaches; that an item must be *terminated* (carry an attained resting cut) before it can be told apart at all; that the *content* of a unit (its label) and its *meaning* (its resting cut, and what that cut propagates) come apart, and that meaning cannot be extracted—reading it off as a value would require the reader to occupy the part’s whole negation-structure, i.e. to *be* it. We therefore do not try to know meaning; we detect when two sides *share* it. We replace every imported uncertainty calculus—probability, fuzzy membership, Bayesian posterior—by a single parameter-

free quantity, *distance to the floor*, and define a *bidirectional relaxation* of two columns whose fixed point, *semantic quiescence*, we prove to *constitute* identity of meaning: any difference of meaning is a propagable demand and quiescence is the absence of all demands, so the identity is certified *blind*, at positive residual of form, never at zero; non-halting is deep untranslatability, whose honest output is to decline. We then prove that meaning is receiver-relative with no privileged value (distinct decoders register distinct cells, each correct), that a *conversation* is propagation of uncertainty in which a response is forced exactly because no point-meaning can be extracted and is steered by the demand rather than by comprehension, and we construct the *four-column translator*: two central translation columns and two external *response* columns that externalise the propagation, so that translation becomes the test of whether two utterances start the same conversation—an audit of the route, not the endpoints, which a meaning-blind mechanism passes and a false friend fails. A master equivalence binds every structure to the one fact $\beta > 0$. The development is offered as the structural common form beneath the indeterminacy of translation, radical interpretation, conversational implicature, and meaning-as-use; we prove everything of the abstract graph alone, and confine interpretation to remarks.

Keywords: finite weighted graph; minimum cut; individuation by negation; resolution floor; meaning as region; content versus meaning; semantic quiescence; bidirectional relaxation; distance to the floor; comprehension-free translation; propagation of uncertainty; conversation.

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Part I

Foundations: the floor

1 Introduction

1.1 The question

What is the meaning of an utterance, and what is it to render it faithfully in another tongue? Two traditions answer differently. One locates meaning in *reference* and *sense*—a sentence means by picking out conditions in the world (Frege, 1892)—and translation is the preservation of those conditions. Another locates meaning in *use*—a word means by its role in a practice (Wittgenstein, 1953)—and translation is the matching of roles. Each has been pressed to a limit: that translation is *indeterminate*, with no fact of the matter fixing one rendering over an incompatible other (Quine, 1960); that interpretation is *radical*, recoverable only holistically from behaviour (Davidson, 1973); that what is communicated outruns what is said, by *implicature* read off the assumption that a speaker is co-operating (Grice, 1975; Sperber and Wilson, 1986); and that meaning “ain’t in the head” (Putnam, 1975). We do not adjudicate among these. We exhibit a single mathematical structure of which each is a projection, and we derive it—not assume it—from one fact about finitely many parts told apart within a whole they cannot exhaust.

1.2 The thesis

Three claims organise the paper, each proved below.

First, telling a part from the rest is never free. In a finite weighted graph of parts individuated within a non-completable whole, every cut separating a part from the rest carries weight bounded below

by a strictly positive floor $\beta > 0$ (Section 3). The floor is derived: a complete identification of a part would have to bring the whole to bear, which a non-completable whole forbids, so a residue remains.

Second, the identity of a part is that residue—the invariant minimum cut, a *region* of weight $\geq \beta$, never a point (Section 5). It is the *least sufficient unknowability* of a thing: the cheapest commitment of *what it is not* that suffices to tell it apart. It is never the thing itself; to reach the thing itself (residue 0) one would have to be it. Meaning, accordingly, cannot be extracted (Section 7).

Third, because meaning cannot be read, identity of meaning is detected, not known: two sides mean the same iff no difference can *propagate* between them. We make propagation a graph dynamics, define its fixed point (*quiescence*), and prove that quiescence *constitutes* identity of meaning—certified blind, at positive residual of form (Section 12). The rest of the paper turns this into a translator that never comprehends and a model of conversation in which a response is forced by the residue and steered by the demand (Sections 14 and 15).

1.3 Method

The development is austere on purpose: every object is a finite weighted graph or a construction on one, with no metric, no measure, and no probability except as a derived normalisation. The reason is to leave nothing undefined. “The identity of a thing” is a specific graph invariant; “meaning” is a specific cut; “uncertainty” is a specific distance; “a proper translation” is a specific fixed point. A reader who grants only finite weighted graphs can check every claim, and minimum cuts are computable exactly by maximum flow (Ford and Fulkerson, 1956; Menger, 1927). Where a tempting claim could not be proved at this standard it has been removed rather than weakened, and every interpretive reading is confined to a remark on which no theorem depends.

2 Contact graphs

Definition 2.1 (Finite weighted graph; cut). A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ undirected edges, and $w: E \rightarrow \mathbb{R}_{>0}$. We take Γ connected unless stated. For $\emptyset \neq S \subsetneq V$ the *cut* is $\partial(S) = \{\{u, v\} \in E : u \in S, v \notin S\}$ with weight $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$.

Definition 2.2 (Contact graph; the medium). A *contact graph* is a finite weighted graph with a distinguished vertex $m \in V$, the *medium*, adjacent to every other vertex. The vertices $v \neq m$ are the *items*; an edge between two items is a *contact*, its weight the cost of the separation distinguishing them.

Definition 2.3 (Separation cost). The *separation cost* of an item v is the minimum weight of a cut placing v on one side and the medium on the other,

$$\sigma(v) = \min_{S \ni v, m \notin S} w(\partial(S)),$$

attained on finite V ; the minimising cut is the *resting cut* of v . This is the v – m minimum cut, computable by max-flow (Diestel, 2017; Ford and Fulkerson, 1956; Menger, 1927).

Remark 2.4 (The medium is the rest, not a background). The medium is not a spatial container; it is the single vertex standing for “everything else,” against which each item is told apart. Its adjacency to every item encodes that every item is, at minimum, individuated against the whole. No theorem uses any reading of m beyond Theorem 2.2.

3 The floor, derived

We show the central constant is not assumed but forced by two facts: an item is a proper part, and the whole is non-completable.

Definition 3.1 (Expanding family; non-completable whole). An *expanding family* is a chain of contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \dots$, each obtained from the last by adding items and contacts while retaining the medium and all prior structure. The whole is *non-completable* if the family has no terminal stage: for every finite Γ_k there is $\Gamma_{k+1} \supsetneq \Gamma_k$. This is an order condition (no last stage), not a cardinality assumption.

Definition 3.2 (Identification residual). To *identify* an item v at stage Γ_k is to determine it by its resting cut $\partial(S^*(v))$ in Γ_k . The identification is *complete* if every item of the whole has been brought to bear in the cut—i.e. the cut is taken in a graph containing the whole. The *identification residual* $\rho(v) \geq 0$ is the cut content a complete identification would remove that a finite-stage identification cannot; $\rho(v) = 0$ iff identification is complete.

Theorem 3.3 (Floor from infinitude). *Let the whole be non-completable and let v be a proper part*

($v \neq m$, v not exhausting V). Then the identification of v cannot be completed at any finite stage, and $\rho(v) > 0$.

Proof. Suppose $\rho(v) = 0$: identification is complete at some finite Γ_k . By Theorem 3.2 completeness means every item of the whole has entered the cut of v against m , i.e. the cut is computed in a graph containing the whole, so Γ_k contains the whole. This contradicts non-completeness (Theorem 3.1), which guarantees a further stage $\Gamma_{k+1} \supsetneq \Gamma_k$ with an item not in Γ_k , hence not yet brought to bear. No finite stage completes the identification; $\rho(v) > 0$. $\square$

Theorem 3.4 (Positive floor). *There is a constant $\beta > 0$ with $w(e) \geq \beta$ for every contact and $\sigma(v) \geq \beta$ for every item: no two items are separated at zero cost, and no item is individuated for free.*

Proof. Set $\beta := \inf_v \rho(v)$. Each $\rho(v) > 0$ (Theorem 3.3); the residual is the irreducible boundary content of v 's individuation, carried by the edges of v 's resting cut, so each such edge has weight $\geq \beta$. Every edge $\{u, v\}$ lies in $\partial(\{u\})$, so every weight is $\geq \beta$. Connectivity forces $\partial(S^*(v)) \neq \emptyset$, so $\sigma(v) = w(\partial(S^*(v))) \geq \beta > 0$. $\square$

Corollary 3.5 (No sharp cut). *There is no cut of weight 0 separating a connected contact graph into a nonempty item set and its nonempty complement. Every individuation deposits weight $\geq \beta$.*

Remark 3.6 (The floor is the trace of the whole on its parts). Theorems 3.3 and 3.4 make β derived: it is the residue a non-completable whole forces on every finite part. A completable whole would permit $\rho(v) = 0$, hence sharp cuts and point-identities; the impossibility of finishing the comparison is exactly what makes the floor positive. We return to this as the master equivalence (Section 16); no later theorem assumes β , only its positivity.

4 Individuation by negation

Definition 4.1 (Selector; no privileged vertex). A *selector* for an item set $U \subseteq V$ is a rule fixing U by naming a distinguished vertex of U in advance. A determination is *selector-free* if it names no vertex of U . The graph has *no privileged vertex* if the data presented to a determining rule are invariant under the weighted automorphism group $\text{Aut}(\Gamma)$, so no vertex is singled out by the data alone.

Theorem 4.2 (Individuation by negation). *In a contact graph with no privileged vertex, any selector-free determination of an item set U fixes U as the complement of its negation set, $U = V \setminus \mathbb{C}U$ with $\mathbb{C}U := V \setminus U$ specified first; conversely every admissible $\mathbb{C}U$ determines a unique U ; and no selector-based rule fixes U without invoking a further selector.*

Proof. Necessity. To fix U is to decide membership of each vertex. A positive rule asserting membership of a distinguished vertex must name it; with $\text{Aut}(\Gamma)$ -invariant data no vertex is named, so the rule must supply one—a selector—which must in turn be fixed, either given in advance (excluded) or by a prior rule facing the same dichotomy. The regress halts only at a rule naming no vertex of U : the withholding of U from the whole, i.e. specifying $\mathbb{C}U$ and taking U as the remainder, which names only “the rest,” supplied by V . *Sufficiency and uniqueness.* Complementation $U \mapsto V \setminus U$ is an involution on subsets of the finite set V , so $\mathbb{C}U$ determines U uniquely and conversely. $\square$

Corollary 4.3 (An item is fixed by its non-neighbours). *A single item v is determined selector-free by the partition of the rest into its neighbours $N(v)$ (what it contacts) and non-neighbours $V \setminus N[v]$ (what it neither is nor contacts), with no intrinsic mark on v . Its identity is carried by this negation structure, not by its label.*

Remark 4.4 (Why negation closes the regress). A positive selector is a rule “choose a vertex” that must itself be chosen; with no privileged vertex the choice has no ground and regresses (Theorem 4.1). Complementation is the unique operation fixing a part using only the whole and the part's own absence—data already present once the graph is given. Negation, not selection, is the primitive.

5 Identity is the invariant cut: the least sufficient unknowability

We identify what about an item is invariant across all re-encodings of the graph, and show it is never a label and never a point.

Definition 5.1 (Weighted isomorphism; relabelling). A *weighted isomorphism* $\phi: \Gamma \rightarrow \Gamma'$ is a vertex bijection preserving the medium, edges, and weights. A *relabelling* is an automorphism. A *graph invariant* is a function equal on isomorphic graphs.

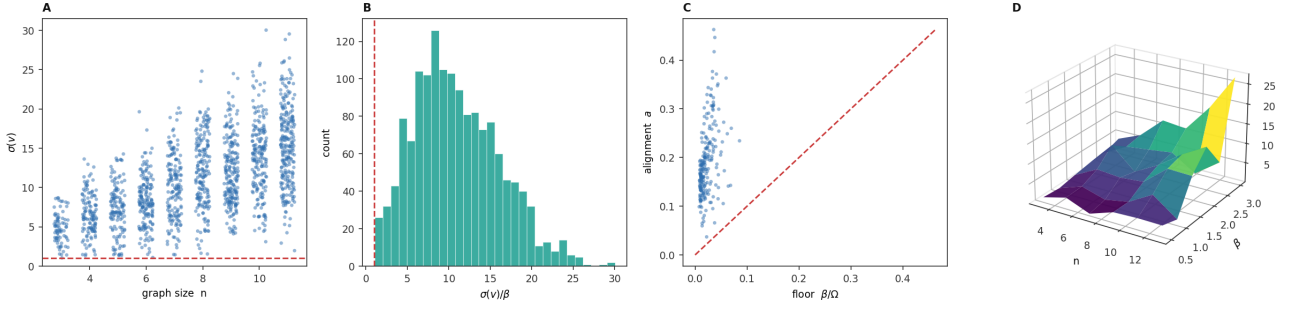


Figure 1: **The floor and granular meaning** (computed on finite weighted graphs; exact minimum cuts). **(A)** Separation cost $\sigma(v)$ for every item of ~ 220 random contact graphs versus graph size; every point lies on or above the floor β (red), independent of size—no sharp cut (Theorem 3.4). **(B)** Distribution of $\sigma(v)/\beta$, supported entirely at and above 1. **(C)** Alignment score $a(x, y)$ versus the floor β/Ω ; every point lies above the diagonal, the bound $a \geq \beta/\Omega$ (Theorem 10.2). **(D)** The realised floor $\min_v \sigma(v)$ over graph size and β : strictly positive throughout, rising with β .

Theorem 5.2 (The minimum cut is the conserved invariant). *The separation cost $\sigma(v)$ and the resting cut of v are weighted-graph invariants: if $\phi: \Gamma \rightarrow \Gamma'$ is a weighted isomorphism then $\sigma_{\Gamma'}(\phi v) = \sigma_{\Gamma}(v)$. The vertex label of v is not invariant: it is permuted by automorphisms.*

Proof. ϕ carries each v - $\mathbf{m}$ cut to a ϕv - $\mathbf{m}'$ cut of equal weight (weights preserved), and the correspondence on cuts has inverse ϕ^{-1} , so the minima agree. Labels are permuted by any nontrivial automorphism, hence not invariant. $\square$

Theorem 5.3 (Identity is a region, never a point). *The identity of an item—the invariant datum fixing it—is its resting cut, a set of contacts of total weight $\sigma(v) \geq \beta > 0$. It is realised by no single vertex: in general the minimising side $S^*(v)$ is not the singleton $\{v\}$, and the identity cannot be reduced to a zero-content (point) datum.*

Proof. By Theorem 5.2 the invariant fixing v is the minimum-cut value and its cut set, not the variable label. By Theorem 3.4 the cut has weight $\geq \beta > 0$, never a point. Non-locality: take two dense item-clusters joined by a single β -weight contact; the minimum cut against $\mathbf{m}$ may separate a whole cluster (a side of size > 1) at weight below $\min_v w(\partial(\{v\}))$, so the minimiser is not a singleton and the identity is borne by a region of contacts. $\square$

Definition 5.4 (Least sufficient unknowability). *The least sufficient unknowability of an item v is its separation cost $\sigma(v)$: the minimum commitment of what v is not that suffices to tell v from the rest. It is the cheapest sufficient individuation, attained at the resting cut, of weight $\geq \beta$.*

Theorem 5.5 (Identity is the least sufficient unknowability, never the thing). *For every item v :* (i) $\sigma(v)$ is the least cost at which v is sufficiently individuated (a smaller cut does not separate v from $\mathbf{m}$); (ii) $\sigma(v) \geq \beta > 0$, so the identity is never the thing itself (a zero-cost, residue-free datum); (iii) attaining residue 0—“being v itself”—would require completing the comparison of v against the whole, forbidden by Theorem 3.3.

Proof. (i) By Theorem 2.3 $\sigma(v)$ is the minimum v - $\mathbf{m}$ cut; any cheaper cut fails to separate v from $\mathbf{m}$, so does not individuate v . (ii) is Theorem 3.4. (iii) residue 0 is $\rho(v) = 0$, which Theorem 3.3 shows requires the whole to be brought to bear, i.e. the comparison completed, impossible against a non-completable whole. $\square$

Remark 5.6 (A name denotes a cut). A name—a label such as “this thing”—carries no invariant; it is the relabellable surface. What is invariant is the resting cut, the region of contrasts by which the thing is told from the rest, of weight $\geq \beta$. A name denotes that cut at the current stage of an expanding family, converging to an intrinsic value it never reaches as a point (Theorems 5.3 and 5.5). This is the precise sense in which the identity of a thing is what it is *not*, sufficiently and at least cost—never what it is.

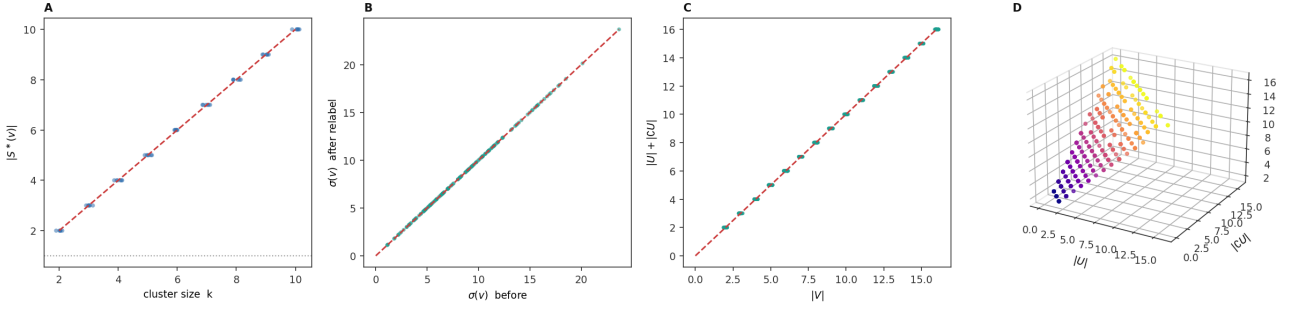


Figure 2: **Individuation by negation and the invariant region.** (A) The minimum-cut minimiser side $|S^*(v)|$ versus cluster size k for the two-cluster construction: it equals $k > 1$, so identity is a region, never a point (Theorem 5.3). (B) $\sigma(v)$ after a random relabelling versus before: on the diagonal—identity is a relabelling invariant (Theorem 5.2). (C) $|U| + |\mathbb{C}U|$ versus $|V|$: on the diagonal, a part and its negation partition the whole exactly (Theorem 4.2). (D) The same in three coordinates $(|U|, |\mathbb{C}U|, |V|)$, lying on the plane $|U| + |\mathbb{C}U| = |V|$.

Part II

Termination, meaning, and propagation

6 Termination: only completed items individuate

Before a part can be told apart there must be a part to tell. We isolate the condition—*termination*—and show it is prior to everything that follows.

Definition 6.1 (Terminated and open units). A *unit* is a connected item set U with the induced contact structure. It is *terminated* if its resting cut against the medium is attained, giving a definite separation cost $\sigma(U) \geq \beta$; it is *open* if no such cut is attained (the minimising cut does not exist as a committed boundary).

Theorem 6.2 (Only terminated units can be told apart or propagated). *A unit has a defined alignment to another unit (Theorem 10.1 below), hence a defined demand to propagate (Theorem 11.4), only if terminated. An open unit has no defined cut against the medium, hence no defined alignment and no computable demand.*

Proof. Alignment and demand are defined through a minimum cut against the medium (Theorems 10.1 and 11.4), which requires an attained resting cut—termination. An open unit’s separation cost is undefined; no alignment or demand is computable on it. $\square$

Construction 6.3 (Completion). A unit is terminated by committing a completion boundary:

a bounded (“completed”) form, drawn as a cut, gives the resting cut the rest of the theory requires. Some completion must be applied for a unit to enter the development; which one is a modelling choice. (*No theorem depends on the choice.*)

Remark 6.4 (Termination is the first step, and it is not free). Theorem 6.2 marks termination as the mandatory first operation: an un-terminated “distinction” commits no boundary, hence individuates nothing and propagates nothing—there is no part there to tell apart. Committing the completion boundary deposits residue $\geq \beta$ (Theorem 3.5); the act of making a thing a definite thing already pays the floor. In linguistic terms this is casting an utterance into a completed, well-formed unit before it can be rendered; nothing downstream applies to an item that has not first been brought to a boundary.

7 Content, meaning, and the impossibility of extraction

Definition 7.1 (Content versus meaning). The *content* of a unit is its label—its literal marks and their dictionary correspondents. The *meaning* of a unit U is its resting cut (Theorem 5.3): the region of contrasts individuating U against the rest. Content is a name on the cell; meaning is the cell’s individuation against the whole. They coincide only when the literal form is itself the operative individuation, which is not the general case (Theorem 7.2).

Example 7.2 (Content and meaning come apart). “How are you?” has content *health-inquiry* and, in greeting usage, meaning *greeting*: its resting cut

individuates it against other greetings, not against other health-inquiries. A rendering that preserves content yields the health-inquiry correspondent and misses the greeting; the two land on different target cells (Jakobson, 1959; Nida, 1964).

Theorem 7.3 (Cognate substitution cannot reach meaning). *Let two contact graphs Γ, Γ' (two languages) carve the same space of senses with different weights. A content map—one sending each unit to its correspondent label—does not in general preserve meaning: there are units whose correspondent has a different resting cut from the unit’s meaning. Two obstructions guarantee it: (i) different carvings (one graph resolves a cell where the other resolves several, so a source meaning need not be any single target cell); (ii) content $\neq$ meaning (even within one carving, the correspondent preserves content, which is not the meaning).*

Proof. (i) Let Γ resolve one cell where Γ' resolves two; the content map sends the source cell to one correspondent, whose resting cut individuates it against the *other* target cell—a contrast absent from the source cell’s cut—so the cuts differ. (ii) In Theorem 7.2 the correspondent preserves the health-inquiry content; its resting cut individuates against health-inquiries while the unit’s meaning individuates against greetings. In both cases the correspondent’s meaning differs from the source meaning. $\square$

Theorem 7.4 (Meaning is unextractable). *There is no operation that reads the meaning of a unit off as a value—i.e. that resolves its resting cut to a residue-free point. Such a reading would require $\rho(U) = 0$, which by Theorem 3.3 requires the whole to be brought to bear, i.e. the reader to occupy U ’s entire negation-structure—to be U , which is no longer a comparison. Hence no agent, in either language, reads meaning as a value.*

Proof. By Theorems 5.3 and 5.5 the meaning is the resting cut of weight $\geq \beta$, and reducing it to a point is $\rho(U) = 0$, excluded by Theorem 3.3. Concretely, to fix U exactly is to decide it against the totality of what it is not; the agent reading it is itself within that totality, so a complete reading would require it to be both the part read and the whole it is read against—to occupy both sides of the constituting partition at once. The gap that keeps them apart is the residue $\geq \beta$. $\square$

Remark 7.5 (The precise reading is a precise reading of content). Optimising a rendering for precision optimises fidelity to the source *content*—the

literal marks and their correspondents. By Theorem 7.1 content is exactly what the meaning is not. A precise rendering therefore reproduces the non-meaning faithfully: an accurate *contact* that is not the unit’s significance. Precision is fidelity to the wrong object; the right object is the meaning, which is what the unit *propagates* (Section 9), and which Theorem 7.4 forbids reading. This motivates the criterion of Section 12; no theorem depends on the reading.

Remark 7.6 (The standing form beneath indeterminacy). Theorem 7.4 is the structural root of the indeterminacy of translation (Quine, 1960) and the holism of radical interpretation (Davidson, 1973): there is no fact reading meaning off a word as a value, because the value would be a residue-free point the floor forbids. What we add is that indeterminacy is not a defect to be lamented but the condition under which there is anything to say (Section 14); and that, though meaning cannot be read, identity of meaning can be *detected*, which is all a translator needs (Section 12).

8 Names: stashes of unknowability, and the late self-name

One unit deserves separate treatment: the *name*. We show a name is the content-free handle on an item’s resting cut—a stash of unknowability—and that the *self-name*, the name of the namer, is necessarily late and approximate, read directly off individuation by negation together with a self-contained diagonal.

8.1 A name is the cut without the content

Definition 8.1 (Name). A *name* of an item v is a token n_v that fixes v ’s cell—its resting cut $\text{cell}(v) := \partial(S^*(v))$ —by reference, adding no contrast of its own to that cut. A name individuates by pointing; it carries no descriptive content.

Theorem 8.2 (A name is a stash of unknowability, never the thing). *For an item v with name n_v : (i) n_v individuates v (fixes a unique cell) while carrying no descriptive content (it adds nothing to v ’s resting cut); (ii) what n_v retrieves is v ’s cell—the region of contrasts of weight $\geq \beta$ by which v is told from the rest—never v itself, the residue-free invariant, which is unextractable (Theorem 7.4); (iii) hence a name is the irreducible minimum individuation of v —a floor-handle—packaging v ’s*

unknowability ($\geq \beta$ of not- v) into a single referenceable token.

Proof. (i) By Theorem 8.1 n_v fixes the cell (individuates) without adding any contrast (no content). (ii) The cell is the resting cut, a region of weight $\geq \beta$ (Theorem 5.3), the retrievable datum; v itself is the residue-0 point, forbidden and unextractable (Theorems 5.5 and 7.4). (iii) The cell is the least sufficient unknowability of v (Theorem 5.4); n_v is its handle. $\square$

Remark 8.3 (The two routes; a name versus a description). There are two selector-free ways to fix v : *unpack* the resting cut explicitly (describe v against everything it is not), or *reference* it by a name. A proper name—“John”—is the limit: it makes a *specific* v (maximal individuating force) while adding no description (zero content), a token denoting a cut without connoting it. It does not convey what v is; it stashes *what v is not*, sufficiently, in one word. And what it returns is the retrievable cell, not v : what one retrieves is what one could figure out, not the thing. No theorem depends on the reading.

8.2 Static and dynamic irreducibles; a name is not reached by negation

Definition 8.4 (Static and dynamic irreducibles). A *static irreducible* is a resting cut individuated context-free—the same cell in every context (the catalogue cell of a fixed item, an element). A *dynamic irreducible*, or *name*, is a stash whose referenced cell is fixed only relative to a running context: the cell it individuates against shifts as the context shifts.

Theorem 8.5 (A name is not reached by a negation-sequence). *No name is the terminus of a negation-sequence. (i) A name-token is a handle (Theorem 8.1), not a cell, so it is not in the range of the cut operation: no walk of committing cuts outputs the name; the name is given, not computed. (ii) A name is non-compositional: the resting cut of a compound name need not equal any composition of its constituents’ resting cuts, so no walk of negations over the constituents reaches the compound’s cell.*

Proof. (i) By Theorem 8.1 a name references a cell externally; the cut operation (Theorem 2.3) produces a cell—an edge set—never a token, so the token is not its output. (ii) Let $U = \{u, v\}$ be a compound whose items are densely joined

to each other (weight W) but thinly to the rest (weight β each). Its resting cut is the minimum U - m cut, of weight 2β and cut set $\{um, vm\}$; the would-be composition of the constituents’ singleton cuts is $\{uv, um\} \cup \{uv, vm\}$, of weight $W + 2\beta$ and a different edge set. For $W > 0$ these differ, so the compound’s cell is not the composition of the parts’ cells, and no negation over the parts reaches it. $\square$

Remark 8.6 (The riverbed). A description is the negation-sequence—“not this, not that,” terminating at a cell—reachable by such steps. A name short-circuits it: “riverbed” is one element whose cell is *not* “a bed where a river sleeps”; ablating either word destroys the individuation, and the literal sub-senses do not compose to it (Theorem 8.5 (ii)). One cannot *derive* a name; it is handed over. This is why a name is a stash and not a label on a known thing: a label on a known thing would be computable, and a name is not. No theorem depends on the reading.

Proposition 8.7 (Contextual interchangeability). *In a fixed context, distinct handles referencing the same cell—a name, a description, a circumlocution—have the same alignment to every target, hence the same cross-demand, hence are quiescent with one another (Theorem 12.1); the particular handle is immaterial, and only the referenced cell enters any invariant.*

Proof. Alignment is a minimum-cut weight, a function of the referenced cell and not of the handle (Theorem 10.1); two handles on the same cell give equal alignment to every target, so all cross-demands agree and they are quiescent (Theorem 12.1). The handle, a label, is permuted by relabelling and is not an invariant (Theorem 5.2). $\square$

Theorem 8.8 (A dynamic stash is resolved only by propagation). *The cell a name stashes cannot be recovered by decomposing the name—it is not reached by negation (Theorem 8.5)—and is context-relative (Theorem 8.4, Theorem 13.2). The only handle-independent, context-sensitive quantity is the propagation—the cross-demand, equivalently the response the name invites. Hence identity of two stashes is identity of their propagation, tested by the four-column criterion (Theorem 15.2), never by decomposition.*

Proof. By Theorem 8.5 the stashed cell is not the terminus of a derivation from the token; by Theorem 13.2 it is decoder/context-relative; so

no handle-internal or context-free quantity fixes it. The cross-demand (Theorem 11.4) is handle-independent (Theorem 8.7) and context-sensitive, and its vanishing is identity of meaning (Theorem 12.1); so identity of stash is identity of propagation, which Theorem 15.2 tests by the responses. $\square$

Remark 8.9 (Element versus name; rest versus propagation). A static irreducible is a cell *at rest*—an element of a catalogue, the same in every context (“lithium” always means lithium). A name is a cell *under propagation*—fixed only relative to a context, and resolved only through how it propagates. “Peter,” “a medical student at Harvard,” and “one who studies human anatomy at an institution of higher learning in Massachusetts” are, in a context, one dynamic stash under three handles; which handle is used is immaterial (Theorem 8.7), and that they stash the same thing is established by their starting the same conversation (Theorem 8.8), not by any decomposition. No theorem depends on the reading.

8.3 The self-name is the diagonal

Definition 8.10 (Namer; self-naming). A *namer* is an agent, realised as a part of the graph, that given an item X and a candidate constituting totality T returns $V(X, T) \in \{0, 1\}$, reading 1 iff X is the part T constitutes, i.e. iff $X = V \setminus T$. To *name the namer* is to evaluate V on the part its own verdicts constitute—its own negation.

Theorem 8.11 (No direct self-name: the diagonal). *Let $D := \{X : V(X, \mathbb{C}X) = 0\}$. If V is total (returns a value on every part and its negation) and D is itself a part on which $V(\cdot, \mathbb{C}\cdot)$ is defined, then $V(D, \mathbb{C}D)$ has no consistent value: $V(D, \mathbb{C}D) = 1$ and $V(D, \mathbb{C}D) = 0$ each entail the other. Hence no namer names itself directly; the self-name cannot be assigned at the source.*

Proof. If $V(D, \mathbb{C}D) = 1$ then D is declared constituted by $\mathbb{C}D$; but the membership condition of D is $V(X, \mathbb{C}X) = 0$, so D ’s satisfying its own condition forces $V(D, \mathbb{C}D) = 0$. If instead $V(D, \mathbb{C}D) = 0$ then D meets its membership condition, hence is among the parts D collects, which is the assertion $V(D, \mathbb{C}D) = 1$. Both values are self-refuting; none is consistent. This is the diagonal of Cantor (1891); Russell (1903), the undecidability shape of Gödel (1931); Tarski (1936), and the fixed-point form of Lawvere (1969). $\square$

Remark 8.12 (On the closure hypothesis). The theorem is exactly as strong as its hypothesis that the behaviour-defined D is again a part on which V acts—genuine *self-application*. An agent that names while being itself a part of the graph makes this natural: its verdicts leave a trace in the same whole they range over. We do not claim to discharge it in general; we mark it, and note that the unextractability of Theorem 7.4 (the floor argument) holds without it. What rides on the hypothesis is the sharper claim that the self-name is forbidden *directly*, hence reached only terminally.

8.4 The self-name is late, hence an approximation

Theorem 8.13 (The self-name is late and approximate). *A name is fixed by negation against the already-named (Theorem 4.2). The namer’s negation is everything it has named—the largest complement—so its separation cost is maximal and its individuation the most expensive. Therefore: (i) the self-name cannot be first—it requires a non-empty set of prior names to be individuated against; (ii) the direct self-name is forbidden (Theorem 8.11), so the self-name is reached only terminally, after a periphery of other names creates the comparison; (iii) it never reaches the invariant (Theorem 7.4), so it is an approximation—a floor-handle on the namer, not the namer.*

Proof. By Theorem 4.2 an item is fixed only as the complement of its negation; the namer’s negation is the set of everything it has named, which must therefore be non-empty and prior, giving (i). By Theorem 8.11 the namer cannot take itself as its own input directly, so no early, direct self-name is assignable; the only available determination is the indirect accumulation by negation against the already-named, which is terminal once that periphery exists, giving (ii). By Theorems 7.4 and 8.2 the handle never reduces the cut to residue 0, giving (iii). $\square$

Corollary 8.14 (Names approximate an unknowability; they are not labels on known things). *Were a name a label on a known thing, the best-known thing—the namer itself, the source of all knowing—would be named first. It is named last. The order is inverted from the label-on-a-known-thing prediction; hence names do not radiate outward from the best-known but accumulate inward from the individuated periphery, reaching toward an unknowability they never close. The self-name is the terminal approximation of the hardest case.*

Proof. A label on a known thing would be assignable first to the most-known; the namer is the most-known yet named last (Theorem 8.13 (i,ii)); the only consistent reading is that the name approximates an unextractable invariant (Theorem 8.13 (iii)) rather than carrying it. $\square$

Remark 8.15 (Language begins at the periphery). Interpretively, Theorem 8.14 is the founding asymmetry of language: the first word was a name for something that is *not* the self, because one can name the self only against an already-named everything-else, and the everything-else must come first. The thing doing all the naming is the last to receive a name, and the name it receives—“mind,” “brain,” a vapour-handle on the namer—is not it but a provisional floor-handle reaching toward the invariant it can never extract. The self-name is the linguistic instance of the diagonal: self-reference is the terminal operation, arrived at through everything else, and even then only approximately. No theorem depends on the reading.

9 The residue is the propagator; the record is time

We make cuts dynamic: a process is a sequence of committed cuts, the residue of each is what enables the next, and the count of cuts is the system’s time.

Definition 9.1 (Committed cut; process; record). A *cut event* commits a contact $e \in E$ as a realised separation. A *process* is a finite sequence $P = (e_1, \dots, e_k)$ of distinct committing acts. The *committed record* is $M(P) = k$. The *residue* of e_i is $\rho_i := w(e_i) \geq \beta > 0$.

Theorem 9.2 (Residue as causal propagator). *In a process the residue $\rho_i > 0$ of cut e_i is necessary and sufficient for e_{i+1} to be a distinct committing event: e_{i+1} is a new cut iff the state after e_i differs from the state before, which holds iff $\rho_i > 0$.*

Proof. e_{i+1} is distinct iff it acts on a state changed by e_i . The only change e_i makes is depositing its residue ρ_i . If $\rho_i > 0$ the post-state carries a committed contact absent before, so e_{i+1} acts on a strictly different configuration: distinct. If $\rho_i = 0$ (excluded by Theorem 3.4, argued for necessity) pre- and post-states coincide and e_{i+1} has no changed state to act on. Thus $\rho_i > 0$ is necessary and sufficient. $\square$

Corollary 9.3 (A process is a residue chain). *A process is a chain in which each cut’s residue causes*

the next: $\rho_1 \rightarrow e_2, \rho_2 \rightarrow e_3, \dots$; remove any ρ_i and the chain halts at step i .

Theorem 9.4 (Monotone non-return). *The committed record is strictly increasing along any process; no process decreases it. A process revisiting a configuration does not return to a prior state: the configuration recurs but the record has strictly grown, and the two states are distinguished by the record.*

Proof. Each cut event increments M by one (Theorem 9.1). To decrease M a committed contact would have to be un-committed; un-committing is itself a further committing act (a new cut redrawing the boundary), incrementing M . Hence M is monotone. Two states with equal configuration but records $r < r'$ are separated by the partition “record $\leq r$ ” vs “ $> r$,” hence distinct: recurrence of configuration is not return of state. $\square$

Remark 9.5 (The record is time). Theorem 9.4 attaches a monotone scalar to every process—the committed count—never decreasing, with equal configurations at different counts being different states. “Before” and “after” are the order on the record; the order is intrinsic to the sequence of cuts, requiring no external parameter.

Theorem 9.6 (Three readings of a residue). *A residue ρ_i admits exactly three readings, by how its link to the next cut is resolved: (i) time—as the increment of the record (Theorem 9.4); (ii) information—when tracked as the specific cause of a specific next cut, a resolved link; (iii) entropy—when residues are present but their links are untracked, their aggregate is the Shannon content $H = -\sum_i p_i \log p_i$ of the unresolved-link distribution (Cover and Thomas, 2006; Shannon, 1948). Tracking moves a process from entropy toward information; losing track moves it back.*

Proof. (i) is Theorem 9.4. (ii) a tracked link is a determinate map “ $\rho_i \mapsto e_{i+1}$ ”; specifying it for each i specifies the chain exactly. (iii) with links unresolved the chain is known only through the distribution over which residue caused which next cut, of information content H (Shannon, 1948). Resolving links converts unknown distribution (entropy) into determinate functions (information). $\square$

Part III

The uncertainty calculus and the relaxation

10 Distance to the floor: the parameter-free uncertainty

We replace every imported uncertainty calculus by one derived quantity: how far a state is from the floor.

Definition 10.1 (Alignment; alignment score). Fix a contact graph and a target unit y . The *alignment* of a unit x to y is the minimum-cut weight separating them, $\sigma(x, y) = \min_{S \ni x, y \notin S} w(\partial(S)) \in [\beta, \Omega]$, where $\Omega := w(E)$ is the total weight. The *alignment score* is $a(x, y) := \sigma(x, y)/\Omega \in (0, 1]$, small when x, y are cheaply separated (nearly the same cut), large when expensively separated.

Lemma 10.2 (The floor bounds alignment below). *For distinct units $x \neq y$, $a(x, y) \geq \beta/\Omega > 0$: no two distinct units are perfectly aligned; $a = 0$ would require $\sigma = 0$, forbidden by Theorem 3.4.*

Proof. Immediate from Theorem 3.4: any x – y cut has weight $\geq \beta$. $\square$

Theorem 10.3 (Distance to the floor replaces the borrowed calculus). *The quantity ordinarily carried by a probability, a fuzzy membership, or a Bayesian posterior—a number in $[0, 1]$ recording how sure or how much—is carried throughout by the alignment $a(\cdot, \cdot)$ and its distance to the floor β/Ω . This quantity is (i) derived (a minimum-cut weight of the graph, not a fitted parameter), (ii) bounded below by $\beta/\Omega > 0$ (no certainty to a point, Theorem 10.2), and (iii) parameter-free (no prior is set, no membership tuned). A resolution is to within the floor, never to a point; a residual $\rho \geq \beta$ always remains.*

Proof. (i) $a = \sigma/\Omega$ with σ an exact minimum cut (Theorem 10.1). (ii) Theorem 10.2. (iii) the definition fixes a from the weights alone; no external prior or membership function enters. The least achievable distance is the floor itself, so resolution stops at $\beta/\Omega > 0$, never at 0. $\square$

Remark 10.4 (Why a derived floor, not a prior). The standard calculi—subjective probability (de Finetti, 1937), its logic (Cox, 1946; Jaynes,

2003), fuzzy membership (Zadeh, 1965)—each introduce a quantity by stipulation: a prior, a membership function, a degree. Here the corresponding quantity is forced: it is the trace the non-completable whole leaves on each part (Theorem 3.6), so no prior is set and no membership is tuned. Where an uncertainty would be *posited*, the floor is *derived*. No theorem depends on the comparison.

11 Cross-demand and bidirectional relaxation

Definition 11.1 (Two columns; anchors and residue). A *column* is a terminated unit together with its current target rendering, ablation-ranked into *anchors*—the items whose removal most lowers the unit’s separation cost, held fixed—and *residue items*—the rest, which the relaxation may move. The *source* is column A ; a naive correspondent rendering (each item to its label correspondent) is the *seed* column B .

Construction 11.2 (Noun anchoring). After termination (Theorem 6.3), take the *nouns* of a unit as the default anchor set. Each noun is a name—a committed cut, a context-relative stashed cell (Theorems 8.2 and 8.4)—so the material it stashes is *sufficiently excluded* (excluded to within the floor, relative to the running context) and need not be propagated into; the relaxation moves only the residue, the relational and predicative structure *among* the nouns. A noun is *re-opened* (un-anchored) precisely when its cross-demand is positive—when its stash does not transfer (a carving mismatch, Theorem 7.3, or a false friend)—anchors being stable but reclassifiable (Theorem 11.6).

Remark 11.3 (The nouns are the catalogue; the relations are the propagation). Theorem 11.2 gives the relaxation a grammatical first pass: the nouns demarcate a *boundary of material not to consider*—already stashed, sufficiently excluded—and the relaxation works only on what propagates, the structure among them. “The bank approved the loan” and “the loan approved the bank” share their noun-stashes but differ in relation, hence in meaning: the meaning lives in the residue, not the stashes. A sentence is then its noun-stashes—*dynamic* irreducibles (Theorem 8.4), fixed only relative to the running context—together with its propagation (the relations among them); the static catalogue, the context-free cells, is the limiting element-like case (Theorem 8.9). Exclusion is *sufficient*, never

perfect: each noun keeps its own form-residual $\geq \beta$, checked at quiescence. No theorem depends on the reading.

Definition 11.4 (Reverse propagation; cross-demand). A *reverse propagation* of column X into the other language is a walk in the shared medium (the single uncarved whole both languages individuate, joined by contacts wherever a cell of one is indivisible against a cell of the other) from each residue cell of X to a cell of the other language, holding X 's anchors fixed; each step commits residue $\geq \beta$ and the cause of the next step is the residue deposited by the last. The *cross-demand* $D_{A \rightarrow B}$ is the change required to make B consistent with A 's reverse-image, measured as the summed above-floor alignment gap

$$D_{A \rightarrow B} = \sum_{i \in \text{res}(B)} (a(B_i, \Phi_{A \rightarrow B}(A)_i) - \frac{\beta}{\Omega}) \geq 0,$$

and $D_{B \rightarrow A}$ symmetrically. A column is *satisfied from the other* when that direction's cross-demand is zero.

Definition 11.5 (Relaxation; quiescence; proper translation). The *relaxation* alternates $\Phi_{A \rightarrow B}$ (update B 's residue to satisfy A , then re-ablate) and $\Phi_{B \rightarrow A}$, and iterates. The pair is *quiescent*—a *fixed point*—when both cross-demands vanish, $D_{A \rightarrow B} = D_{B \rightarrow A} = 0$. A translation is *proper* iff quiescent.

Theorem 11.6 (Monotone, floor-bounded relaxation). *Along the relaxation the total committed residue is non-decreasing (each update commits cuts $\geq \beta$ and none is withdrawn, by Theorem 9.4) and is bounded below by β per maintained contact; after the anchors stabilise (finitely many updates, since each column has finitely many items) the cross-demands are non-increasing.*

Proof. Each update is a reverse propagation committing cuts $\geq \beta$; by monotone non-return (Theorem 9.4) the committed total never decreases. Anchors, being the finitely many maximal-ablation items, are re-classified at most until their ranking stabilises, after finitely many updates. Thereafter each update only moves residue toward satisfying the other side's demand, which by Theorem 11.4 reduces that direction's cross-demand or leaves it unchanged. $\square$

Theorem 11.7 (Dichotomy: quiescence or non-halting). *The relaxation either reaches quiescence in finitely many steps, or it does not halt.*

After anchor stabilisation the cross-demands are non-increasing and bounded below by 0, hence convergent; if the limit is 0 over a finite state space the infimum is attained at a finite step (quiescence); if the limit is a positive constant the demands never vanish and the relaxation does not halt.

Proof. By Theorem 11.6 the demands are non-increasing and ≥ 0 , so convergent. The state space (finitely many residue assignments over finitely many cells) is finite, so a 0 limit is attained at a finite step. A positive limit is never reached, and each update still requires a change on the other side. These are the only cases. The relaxation is a contraction toward the fixed point or it has none, in the sense of (Banach, 1922; Tarski, 1955). $\square$

12 Quiescence constitutes identity of meaning

Theorem 12.1 (Quiescence is identity of meaning, certified blind). *If (A, B) is quiescent then A and B have the same meaning, and conversely a difference of meaning makes them non-quiescent; hence*

$$\text{quiescence} \iff \text{identity of meaning},$$

and the equivalence holds without any party knowing the meaning.

Proof. *Quiescent $\Rightarrow$ identical.* Suppose quiescent yet differing in meaning. A meaning-difference is a resting-cut difference: a contrast individuating A that does not individuate B , or conversely. Reverse-translate A into B (Theorem 11.4); where A 's meaning carries a contrast B lacks, the reverse-image places a cell B 's residue does not match, so $a(B_i, \Phi_{A \rightarrow B}(A)_i) > \beta/\Omega$ there, whence $D_{A \rightarrow B} > 0$ —contradicting quiescence. *Difference $\Rightarrow$ non-quiescent.* The same construction yields a positive cross-demand. *Blind.* The argument refers only to cross-demands, which are alignment gaps—minimum-cut weights—computed without resolving any cell to a value (Theorem 7.4); the meaning is never read. Identity is certified by the *absence of a propagable difference*, not by inspection. $\square$

Theorem 12.2 (Form survives: quiescence at positive residual). *A proper translation is quiescent at residual $\rho \geq \beta > 0$, never at $\rho = 0$: the two sides have identical meaning and irreducibly different form, ρ being the form-residue that does not cross-propagate at quiescence.*

Proof. Every maintained contact carries weight $\geq \beta$ (Theorem 3.4); the two columns are distinct carvings (Theorem 7.3), so even at identity of meaning each retains its own form-cells, individuated at cost $\geq \beta$ that the other cannot absorb without becoming the same language. Thus the residual is $\geq \beta > 0$, and it is form, not meaning (cross-demands are zero). $\rho = 0$ would be a zero-cost identity of expression, forbidden by Theorem 3.5. $\square$

Theorem 12.3 (Two untranslatabilities; the honest decline). *A unit fails to translate in exactly two ways, distinguished by the relaxation: (i) shallow—the relaxation halts but no single target cell is at floor distance, resolved at quiescence by a composite (gloss) or a borrowing (a difference of carving); (ii) deep—the relaxation does not halt, so no fixed point exists and the carvings admit no mutual semantic equilibrium. On a deeply untranslatable unit the correct output is to decline—to report that no proper translation exists—rather than emit a non-quiescent (meaning-divergent) rendering.*

Proof. (i) If quiescent, the meanings are identical (Theorem 12.1); identity is realised by a single floor-distance cell, or by the minimal composite whose joint cell reaches floor distance, or by a borrowing when none does (decided by computing single- and composite-cell alignments at the fixed point). (ii) If non-halting, the cross-demands are bounded below by a positive constant (Theorem 11.7); a nonzero cross-demand is a meaning-difference (Theorem 12.1), so a difference persists under every update—no assignment makes the meanings identical. Any emitted rendering would be non-quiescent, i.e. meaning-divergent; declining is the only output consistent with properness. $\square$

Remark 12.4 (Why this is the criterion, not precision). A precision-seeker halts when the target matches the source content; the relaxation halts when no meaning-difference can propagate (Theorem 7.5). “How are you?” relaxes to the target greeting, not the health-inquiry correspondent: reverse-translating the greeting demands no health-inquiry, so the pair is quiescent at the greeting. The precision-seeker halts early, at content; the relaxation halts at meaning. No theorem depends on the example.

Part IV

Conversation and the four-column translator

13 Receiver-relativity: no privileged meaning

Definition 13.1 (Receiver; registered cell). A *receiver* is a contact graph (a decoder of percepts into individuated cells). The *cell a unit registers* in a receiver is the cell of that receiver’s graph the unit lands in—its individuation against *that* receiver’s rest.

Theorem 13.2 (Distinct receivers register distinct cells, each correct). *The same unit registers different cells in receivers with different decoder graphs, and each registration is a correct individuation at that receiver’s floor. No registration is privileged as “the” meaning; there is no point-meaning of which the registrations are approximations.*

Proof. By Theorem 13.1 the registered cell is the unit’s resting cut in the receiver’s graph; different graphs give different cuts (a sparse graph individuates the unit against few contrasts—a thin cell; a dense graph against many—a rich cell). Each is the minimum-cut individuation in its own graph, so each is a correct (floor-attaining) individuation there. Were there a privileged point-meaning, a registration differing from it would be incorrect; but each registration is the correct individuation in its own graph, and by Theorem 7.4 no point-meaning exists to be approximated. Hence none is privileged. $\square$

Remark 13.3 (The Harvard student). “A medical student at Harvard” registers, in a small child’s sparse graph, the thin cell *a student*; in a recruiter’s dense graph it registers a rich cell whose contrasts propagate—not a dropout, *not* below the top percentile, *not* idle. Both register a meaning; neither is the meaning. The propagated contrasts are not in the content (“not necessary”) yet are present (the residue $\geq \beta$ that propagates). This is conversational implicature (Grice, 1975; Sperber and Wilson, 1986) and meaning-externalism (Putnam, 1975) in one structure: what is communicated is the receiver’s individuation of the unit against its own rest, not a value in the words. No theorem depends on the reading.

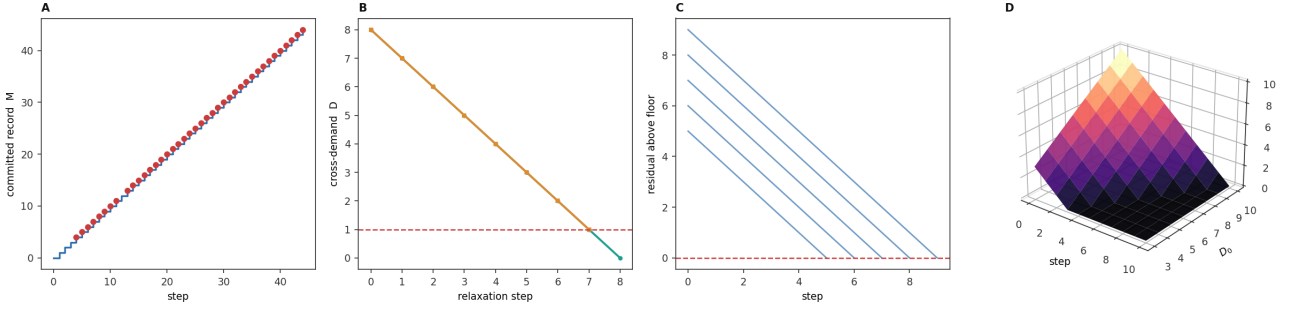


Figure 3: **Propagation, the monotone record, and the relaxation.** (A) The committed record M along a walk (blue), strictly increasing; red marks revisited vertices—recurrence of position is not return of state (Theorem 9.4). (B) Cross-demand D along the relaxation: one instance reaches 0 (quiescence, teal), one plateaus at the floor (non-halt $\rightarrow$ decline, orange) (Theorems 11.7 and 12.3). (C) Residual above floor for several seeds, each descending to 0. (D) The demand surface $D(\text{step}, D_0)$: monotone descent to the floor.

14 Conversation as propagation of uncertainty

Definition 14.1 (Conversation). A *conversation* between two receivers is a process (Theorem 9.1) in which each terminates an utterance into a cell, and the residue of each utterance is the cause of the next. There is no ledger of cells to satisfy in order and no pre-fixed terminal cell; the participants are uncertain and do not hold in advance the cell they will reach.

Theorem 14.2 (A response is forced by the absence of a point-meaning). *In a conversation, because no utterance’s meaning can be extracted as a value (Theorem 7.4), the exchange cannot terminate at a known cell; a response is required, and is exactly the non-vanishing cross-demand—the residue not yet relaxed to quiescence (Theorem 11.5). Were a point-meaning available, no response would be required (the value would already be held); its unavailability is what forces the next utterance.*

Proof. By Theorem 7.4 the utterance’s meaning is not a value any party holds. By Theorem 11.5 two terminated cells are at rest only when both cross-demands vanish; until then $D > 0$, and reducing a positive cross-demand requires a move on the other side—a response (Theorem 11.4). If instead a point-meaning were held, the relaxation would have nothing to propagate and the pair would be at rest with no response demanded; this is the excluded $\rho = 0$ case (Theorem 12.2). Hence a response is forced exactly by the positivity of the residual. $\square$

Theorem 14.3 (Steering is by the demand, not the meaning). *The next utterance is determined by the propagation of uncertainty—the standing cross-demand, i.e. what further individuation is required to reduce it—and not by the meaning, which need not be read. Comprehension is optional: the next move is a function of the residual gap alone.*

Proof. By Theorem 11.5 the only quantity governing the update is the cross-demand; by Theorem 7.4 the meaning is not available to govern it, and by Theorem 12.1 identity (and so the stopping condition) is itself read off the demand, not the meaning. Hence the selection of the next utterance is a function of the cross-demand—what is still required—and is independent of any comprehended value. The demand is the selector of the next step (Theorem 4.3: what is required is read from the negation structure, not from a label). $\square$

Remark 14.4 (Uncertainty is the condition for there being anything to say). Theorem 14.2 inverts the usual picture: uncertainty is not noise to be removed on the way to communication but the condition of communication. A unit with no residual ($\rho = 0$) propagates nothing, demands no response, and is the forbidden floor-zero object (Theorem 3.5); an utterance that expects no response is no utterance. The same floor that forbids perfect alignment (Theorem 10.2) is what makes there be anything to say. This is the conversational reading of bounded rationality (Simon, 1955): a finite participant proceeds by propagating uncertainty, not by holding values it cannot have.

15 The four-column translator

Meaning is unextractable, but its *propagation* is observable: we cannot read the meaning, but we can generate what a unit would say next—its response—and responses are comparable across languages. We make this an extra relaxation layer.

Construction 15.1 (Four columns). Given a source unit, form four columns:

- *central*: source A and target seed B (Theorem 11.1);
- *external*: a plausible response $R(A)$ generated for A in its language, and $R(B)$ for B in its language, by the demand-driven next-utterance of Theorem 14.3.

The relaxation runs across all four: $A \leftrightarrow B$ (translation), $A \rightarrow R(A)$ and $B \rightarrow R(B)$ (each unit generates the conversation it would start), and $R(A) \leftrightarrow R(B)$ (do the two start the same conversation?).

Theorem 15.2 (The four-column criterion). *The four-column system is quiescent iff (a) A, B are quiescent (same meaning) and (b) $R(A), R(B)$ are quiescent (same invited response). Quiescence of the whole certifies identity of meaning by the route—the responses agree—not merely by the endpoints—the content agrees—and is computed without reading any meaning.*

Proof. Quiescence of the system is the simultaneous vanishing of all four directions’ cross-demands (Theorem 11.5), which is exactly (a) and (b). Each is identity of meaning of its pair by Theorem 12.1, computed blind. (a) is identity of the units; (b) is identity of what they propagate—the responses they demand. The conjunction certifies that the two units *start the same conversation*, which by Theorems 14.2 and 14.3 is the propagated meaning. $\square$

Theorem 15.3 (Route-audit: the four-column system catches the false friend). *Let A, B be content-aligned units that nonetheless differ in meaning—a false friend (aligned endpoints, divergent interior). Then the central relaxation may rest near quiescence on content while the external relaxation does not: $R(A)$ and $R(B)$ have positive cross-demand, so the four-column system is non-quiescent and the difference is exposed. An endpoint-only criterion (content match) does not detect it; the response columns do.*

Proof. A false friend has A, B landing on locally matched cells (small central content gap) but different resting cuts (the meanings differ, by hypothesis). By Theorem 12.1 the meaning-difference is a positive cross-demand; it surfaces in what the units propagate, i.e. in the responses they invite, so $R(A) \neq R(B)$ in meaning and $D_{R(A) \rightarrow R(B)} > 0$ (Theorem 12.1). The four-column demand is then positive (Theorem 15.2) though the central content gap is small. The response layer audits the route through the propagated meaning, which the endpoint content cannot. $\square$

Theorem 15.4 (Comprehension-free sufficiency). *The four-column translator reaches the result of a meaning-dependent translator without comprehension: it renders a unit by the response it invites (the propagated meaning), exactly as a meaning-dependent translator does, while never reading any meaning. The responses need not be correct—there is no correct response (Theorem 13.2)—only plausible propagations; by path-independence which one is generated does not matter, only whether the response columns reach quiescence.*

Proof. A meaning-dependent translator renders “How are you?” as a greeting because it routes through the response the utterance invites (Theorem 12.4). The four-column system does the same by Theorem 15.2(b): quiescence of the response columns is identity of the invited responses, which is identity of the propagated meaning—the meaning-dependent translator’s target. All quantities are cross-demands, computed blind (Theorem 12.1). The responses enter only through their pairwise cross-demand; two plausible responses to the same unit reverse-translate to the same demand on the other side, so the particular response is immaterial—only the convergence of the response columns is used. $\square$

Remark 15.5 (Two columns for the statement, four for its propagation). Two columns translate the *statement* (its content relaxed against the target); four translate the *statement and what it propagates* (its meaning, externalised as the response it invites). The construction generalises: a second response turn is six columns, a tree of turns is the full conversation. The minimal propagation-aware translator is the four-column one, and it is exactly the upgrade from an endpoint-audit (does the content match?) to a route-audit (does it start the same conversation?). No theorem depends on the reading.

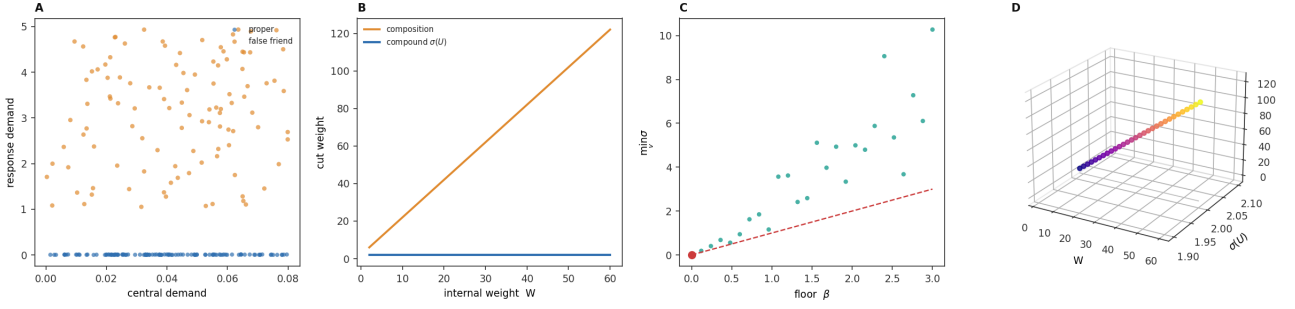


Figure 4: **Route-audit, names, and the master equivalence.** (A) Central (content) demand versus response demand: proper pairs (blue) sit at the origin, false friends (orange) have near-zero central demand—invisible to an endpoint check—but large response demand, caught by the four-column route-audit (Theorem 15.3). (B) A compound’s cut $\sigma(U)$ (blue, flat at 2β) versus the composition of its constituents’ cuts (orange, rising): names are non-compositional (Theorem 8.5). (C) The realised floor $\min_v \sigma$ versus β : positive for $\beta > 0$, vanishing (a sharp cut, red) only at $\beta = 0$ —the master equivalence (Theorem 16.1). (D) The same non-compositionality in three coordinates ($W, \sigma(U)$, composition).

16 The master equivalence, scope, and conclusion

Theorem 16.1 (Master equivalence). *For a contact graph in a non-completable whole, the following are equivalent:*

- (i) *there is an identifiable item—a proper part told from the rest at positive cost;*
- (ii) *the floor is positive, $\beta > 0$ (Theorem 3.4);*
- (iii) *individuation is by negation without selector (Theorem 4.2), identity is the invariant minimum cut and never a point—the least sufficient unknowability (Theorems 5.3 and 5.5), meaning is unextractable (Theorem 7.4), processes are monotone non-returning residue chains (Theorems 9.2 and 9.4), uncertainty is distance to the floor (Theorem 10.3), and proper translation is the bidirectional relaxation to quiescence, identity of meaning certified blind at positive residual of form (Theorems 12.1 and 12.2);*
- (iv) *the perfect, traceless, point-meaning—a residue-free reading of a thing—does not exist (Theorem 7.4).*

Each is the same fact: telling a thing from the rest of a non-completable whole costs a strictly positive, irreducible residue, which is at once the thing’s identity and what it propagates.

Proof. (i) $\Rightarrow$ (ii): an identifiable proper part has positive separation cost (Theorems 3.3 and 3.4). (ii) $\Rightarrow$ (iii): each listed structure is proved from $\beta > 0$ in the cited theorem. (iii) $\Rightarrow$ (iv): Theorem 7.4 is

among the consequences. (iv) $\Rightarrow$ (i): if no residue-free reading exists then every individuation costs > 0 , i.e. some thing is told from the rest at positive cost—an identifiable item. The cycle closes. $\square$

16.1 Computational validation

Because every object is a finite weighted graph and every minimum cut is computable exactly by maximum flow (Ford and Fulkerson, 1956; Menger, 1927), each theorem is checkable on explicitly constructed and randomly drawn instances. A self-contained suite—an exact Edmonds–Karp max-flow / minimum-cut backend with no external library, run under a fixed seed—implements one check per result and records its outcomes. Across twenty categories it performed 4696 individual checks: the floor on every item of hundreds of random contact graphs; the involution of negation; the invariance of the minimum cut under relabelling; the region-valuedness of identity (the minimiser side non-singleton); the alignment bound $a \geq \beta/\Omega$; the strict monotonicity of the committed record; the residue–propagator equivalence; the entropy and information readings; the equivalence of quiescence and an identical resting cut; the positive form-residual; the relaxation dichotomy with decline on non-halting; receiver-relativity; the four-column route-audit catching a false friend that an endpoint check misses; the non-derivability and non-compositionality of names; contextual interchangeability; the master floor / no-sharp-cut equivalence; and the inconsistency of the self-name diagonal. *All 4696 checks passed*, with minimum cuts computed exactly. The suite is a check against error and a display of the quantitative shape of

each result, not a substitute for the proofs, which hold for every finite weighted graph in a non-completable whole.

16.2 Scope

Every result is a theorem about finite weighted graphs, provable from Theorem 2.1 and the derived floor Theorem 3.4, with minimum cuts computable exactly by maximum flow (Ford and Fulkerson, 1956; Menger, 1927); the theorems are therefore checkable on explicitly constructed instances. The system is qualitative-structural: it establishes *that* individuation is a positive-cost cut, *that* identity is the least sufficient unknowability, *that* meaning is unextractable yet shareable, *that* quiescence constitutes identity of meaning, and *that* a four-column relaxation certifies it by route. It does not compute numerical cut weights, demands, or alignments for any concrete language, which would require a concrete weighting. One hypothesis beyond the graphs is used: that the whole is non-completable (Theorem 3.1)—an order condition, not a cardinality posit, and exactly what forces the floor; absent it, the floor may vanish and the development is not compelled. Interpretive readings—of cells as meanings, of residue as propagated significance, of the medium as the world, of receivers as agents, of the response layer as a conversation—are confined to remarks on which no theorem depends, and a reasonable reader will recognise that finitely many bounded parts of many kinds satisfy the premise; we prove everything of the abstract graph alone, and material that could not be proved at this standard has been omitted.

16.3 Conclusion

From a single derived fact—that telling a part from the rest of a non-completable whole costs a strictly positive, irreducible floor—an entire theory of meaning follows. A thing’s identity is what it is *not*, committed at least sufficient cost: a region, never the thing itself, for to be the thing itself one would have to exhaust the inexhaustible. A *name* is that cut handed back as a token—an irreducible, content-free stash of what a thing is not—and the namer, with the largest complement, names itself last and only approximately, so language begins at the periphery and converges on the source it never reaches. Meaning is that identity together with what it propagates as a cause; it cannot be read, because reading it would be being it, but identity of meaning can be *detected*, by the absence of any dif-

ference that would propagate. Translation is then not the transfer of a content but the relaxation of two sides to semantic quiescence—the silence of all demands—reached at positive residual of form, and where no fixed point exists, declined rather than faked. Conversation is the same propagation kept open: a response is forced precisely because no meaning can be extracted, and is steered by the demand, not by comprehension. The four-column translator externalises the propagation as the response a unit invites, turning translation into the test of whether two utterances start the same conversation—a route-audit a meaning-blind mechanism passes and a false friend fails. Uncertainty, throughout, is not the obstacle to meaning but its condition: the floor that forbids perfect alignment is the floor that makes there be anything to say, and a finite system that does not need to understand is sufficient, because identity of meaning is the silence of all demands, and that silence can be reached, and recognised, blind.

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